Tushar Kumar Barman

India | +91 9613847976 | tusharbarman123@gmail.com
linkedin.com/in/tusharbarman | github.com/tusharkrbarman

Professional Summary

Software Engineer with 2+ years in **DevOps** and **cloud platform automation** at Intel, focused on CI/CD automation, AWS/Terraform infrastructure, observability, and internal developer tooling. Engineered self-service provisioning and golden-image automation, deployed Kubernetes-based platform services, and delivered observability dashboards for reliable CI validation at scale.

Experience

Software Development Engineer | DevOps and Platform Engineering — Jul 2024 - Feb 2026
Intel Corporation — *Hyderabad, India*

- Enabled **self-service Jenkins validation** across internal clouds by building a provisioning API for machine reservation, OS deployment, and status tracking; deployed on **AWS ECS Fargate** with Terraform-managed ALB, DynamoDB, CloudWatch, and GitHub Actions CD.
- Unified CI machine-utilization and reservation telemetry from Intel OneCloud and GTAX by building **GitHub Actions collectors** that normalized provider API responses and published validated records to **Elasticsearch.**
- Improved capacity planning and issue detection across **1000+ internal cloud systems** with **Grafana SLA/SLO dashboards** for machine utilization, GitHub-related failures, and dirty-image validation waste.
- Protected Intel oneAPI release quality with **CI validation pipelines** integrating cross-team components, acceptance tests, Coverity/BDBA security scans, and market-readiness qualification.
- Reduced environment drift and validation failures with **Packer-managed Windows/Linux golden images,** AWS EC2-built WIMs, Artifactory versioning, and repeatable OneCloud/PXE deployment.
- Validated scalable CPU inference on a **private AWS EKS platform prototype** using Terraform, ALB, FastAPI, OpenVINO/OVMS, readiness probes, metrics-server, HPA, and automated checks.

Infrastructure and DevOps Intern — Jan 2024 - Jun 2024
Intel Corporation — *Hyderabad, India*

- Improved CI capacity and reliability decisions by resolving data-quality, visualization, and reporting-script issues in production **observability dashboards.**
- Improved AIOps failure-detection quality by classifying **1M+ log lines** into curated labeled datasets for build and validation infrastructure.

Education

ME in Computer Science — Aug 2022 - Jun 2024
BITS Pilani — *GPA: 8.95, top 10 in class*

BE in Computer Science — Aug 2017 - Sep 2021
Assam Engineering College — *76.55% (First-Class Hons.)*

Projects

AI Spam Call Screening Platform
Python, LiveKit, Sarvam AI, OpenAI API, Telegram, Render, STT, TTS

- Built an **AI call-screening agent** that answers suspected spam calls, transcribes and classifies the conversation, sends the spam summary through messaging alerts, and prepares TRAI DND complaint reporting after user approval.

Publications

Does Varying BeeGFS Configuration Affect the I/O Performance of HPC Workloads? — IEEE Cluster 2023
doi.org/10.1109/CLUSTERWorkshops61457.2023.00010

- Provisioned a **10-node bare-metal BeeGFS cluster** and automated management, metadata, storage, and client node configuration using **Ansible**; benchmarked I/O throughput using IOR, DLIO, and HACC-IO.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, Bash
DevOps/Cloud: AWS (EC2, ECS Fargate, ECR, ALB, IAM, VPC, CloudWatch), Terraform, GitHub Actions, Jenkins, Docker, Kubernetes, Artifactory, CI/CD, Linux
Backend/Data/AI: REST APIs, FastAPI, NestJS, DynamoDB, Elasticsearch, OpenVINO/OVMS